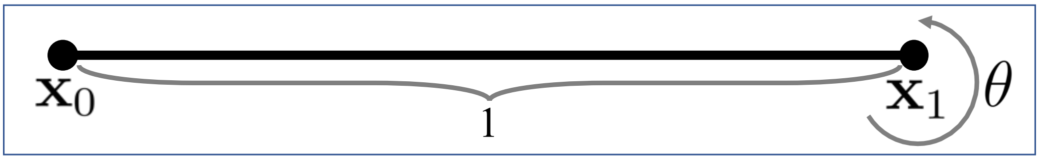
— *Solution notes* —

COMPUTER SCIENCE TRIPOS Part IB – 2024 – Paper 7

7 Further Graphics (aco41)

Quaternions

(*a*) You are given a unit length rod as in the figure. There is no rotation at one end of the rod at point $\mathbf{x}_0$ and a rotation around the plane normal $\mathbf{n}$ is defined at the other end $\mathbf{x}_1$. This rotation is interpolated along the rod. The interpolation weight for $\mathbf{x}$ on the rod is $||\mathbf{x} - \mathbf{x}_i||$ for the rotation at $\mathbf{x}$, $i = 0, 1$.



(*i*) Write the rotation in quaternion form at $\mathbf{x}$ given a rotation of $\theta \leq \pi$ at $\mathbf{x}_1$ assuming shortest path interpolation of rotations in SO(3). [3 marks]

(*ii*) At $\mathbf{x}_1$ at time $t = 0$ there is no rotation and at $t = 1$ the rotation angle is $\pi/2$. Write the quaternion at $\mathbf{x}$ at any time in $[0, 1]$ assuming the shortest path interpolation in SO(3) over time and over the rod. [3 marks]

(*iii*) Answer (*ii*), this time assuming linear blending of quaternions along the rod and the shortest path interpolation in SO(3) over time. [3 marks]

(*iv*) Write an expression for the norm of the quaternion in (*iii*). [3 marks]

Answer:

(*i*) Let $l(\mathbf{x}) = ||\mathbf{x} - \mathbf{x}_0||$ then $\mathbf{q}(\mathbf{x}) = \cos(\theta l(\mathbf{x})/2) + \mathbf{n}_q \sin(\theta l(\mathbf{x})/2)$ where $\mathbf{n}_q$ is in pure quaternion form $\mathbf{n}_q = in_x + jn_y + kn_z$. An answer with $\mathbf{n}_q = -(in_x + jn_y + kn_z)$ is also correct.

(*ii*) $\mathbf{q}(l(\mathbf{x}), t) = \cos(\pi l(\mathbf{x})t/4) + \mathbf{n}_q \sin(\pi l(\mathbf{x})t/4)$.

(*iii*) At time t, the angle of rotation at $\mathbf{x}_1$ is $\pi t/2$ and the resulting quaternion is $\mathbf{q}(\mathbf{x}_1, t) = \cos(\pi t/4) + \mathbf{n}_q \sin(\pi t/4)$. The quaternion at $\mathbf{x}_0$ is always given by $\mathbf{q}(\mathbf{x}_0, t) = 1$. Then we have $\mathbf{q}(\mathbf{x}, t) = [\cos(\pi t/4) + \mathbf{n}_q \sin(\pi t/4)]l(\mathbf{x}) + 1 - l(\mathbf{x})$. We could also first do the linear blending of quaternions along the rod and then the shortest path blending over time. This would result in: $\mathbf{q}(\mathbf{x}, t) = ([\cos(\pi/4) + \mathbf{n}_q \sin(\pi/4)]l(\mathbf{x}) + 1 - l(\mathbf{x}))^t$.

(*iv*) For the case of interpolating over time first, we can group the terms as $\mathbf{q}(\mathbf{x}, t) = [\cos(\pi t/4) + \mathbf{n}_q \sin(\pi t/4)]l(\mathbf{x}) + 1 - l(\mathbf{x}) = [\cos(\pi t/4)l(\mathbf{x}) + 1 - l(\mathbf{x})] + \mathbf{n}_q \sin(\pi t/4)l(\mathbf{x})$. Then the norm is $\sqrt{a^2 + b^2}$ where $a = \cos(\pi t/4)l(\mathbf{x}) + 1 - l(\mathbf{x})$ and $b = \sin(\pi t/4)l(\mathbf{x})$. For the case of interpolating along the rod first, we can use the equivalent definition of the norm with $\mathbf{q}\mathbf{q}^*$.

Representing the plane of a triangle

(*b*) Given a triangle in 3D with vertex locations $\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3$,

(*i*) determine a condition on the vertices for the triangle to define a valid plane, [1 mark]

(*ii*) define a parametric form $\mathbf{p}(u, v)$ for the plane of the triangle assuming it defines a valid plane, [2 marks]

(*iii*) write an expression for the normal of the plane, [1 mark] *(iv)* write the steps of an algorithm to find the closest point of a point $\mathbf{x}$ in space on the triangle. [4 marks]

Answer:

(i) The difference vector between any two vertices, e.g. $\mathbf{v}_3 - \mathbf{v}_1$ and $\mathbf{v}_2 - \mathbf{v}_1$, should be linearly independent.

(ii) One possible parametrisation is $\mathbf{p}(u, v) = \mathbf{v}_1 + u(\mathbf{v}_3 - \mathbf{v}_1) + v(\mathbf{v}_2 - \mathbf{v}_1)$.

(iii) The plane normal can be written as $\mathbf{n} = (\mathbf{d}_{3,1} \times \mathbf{d}_{2,1})/||(\mathbf{d}_{3,1} \times \mathbf{d}_{2,1})||$, where $\mathbf{d}_{3,1} = \mathbf{v}_3 - \mathbf{v}_1$ and $\mathbf{d}_{2,1} = \mathbf{v}_2 - \mathbf{v}_1$. [2 marks] The normal is defined up to sign. [1 mark]

(iv) The closest point can be on the triangle, on an edge, or one of the vertices of the triangle. A possible algorithm is: 1- find the projection $\mathbf{x}'$ of $\mathbf{x}$ on the plane defined by the triangle, 2 - check if $\mathbf{x}'$ is on the triangle and if so, return $\mathbf{x}'$, 3 - project $\mathbf{x}'$ onto each of the edges and compute the distance to each, 4 - return the closest edge point (this could be a vertex if the closest point on the edge corresponds to a vertex).
